Copy number-aware methylation deconvolution analysis of cancers using whole genome bisulfite sequencing data

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Introduction

Aberrant DNA methylation is a feature of metastatic castration resistant prostate cancer (mCRPC) [1]. Whole genome bisulfite sequencing (WGBS) of bulk tumors reveals DNA methylation at base-pair resolution, however tumor methylation signals are confounded by varying purity and ploidy. We developed CAMDAC to address these confounders [2] and present an extention of this method for WGBS analysis. Applied to mCRPC samples, CAMDAC purifies bulk tumor methylation and offers new opportunities to interrogate epigenetic disruption in cancer evolution.

Method

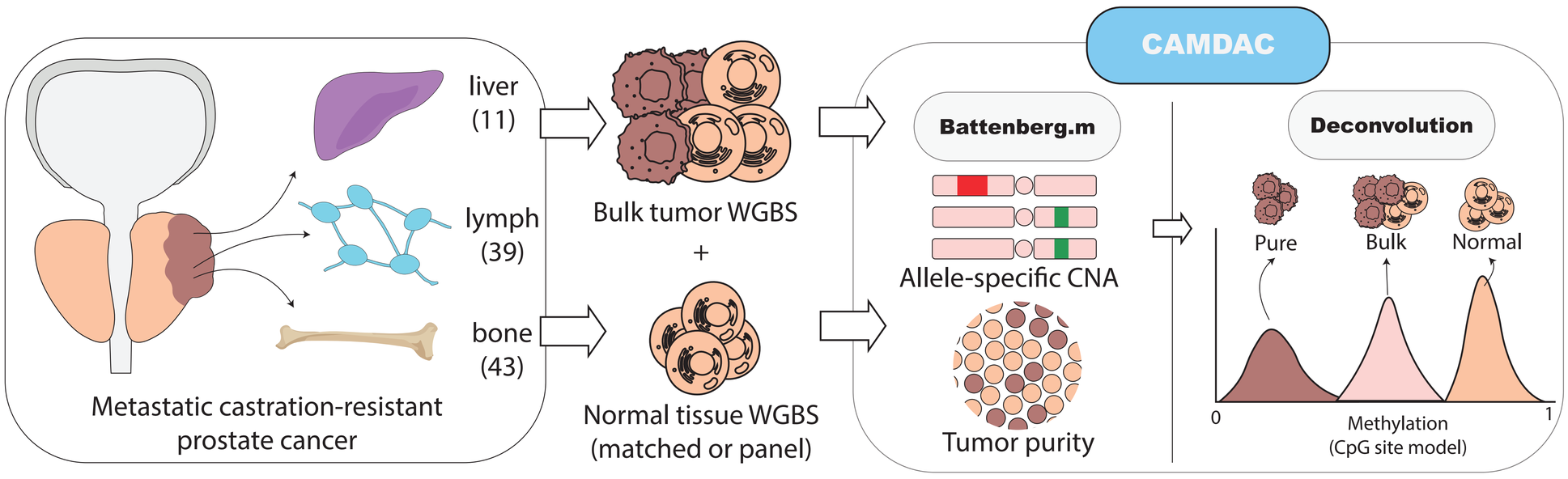

Figure 1. Overview of the CAMDAC workflow. WGBS data was acquired from 93 patients with mCRPC enrolled on the Stand Up To Cancer project [1]. Bismark-aligned BAMs were processed using CAMDAC. Tumor purity and ploidy were estimated using the Battenberg algorithm, followed by deconvolution using the CAMDAC algorithm to estimate pure-tumor methylation at each CpG site.

CAMDAC detects allele-specific copy number abberations concordant with WGS

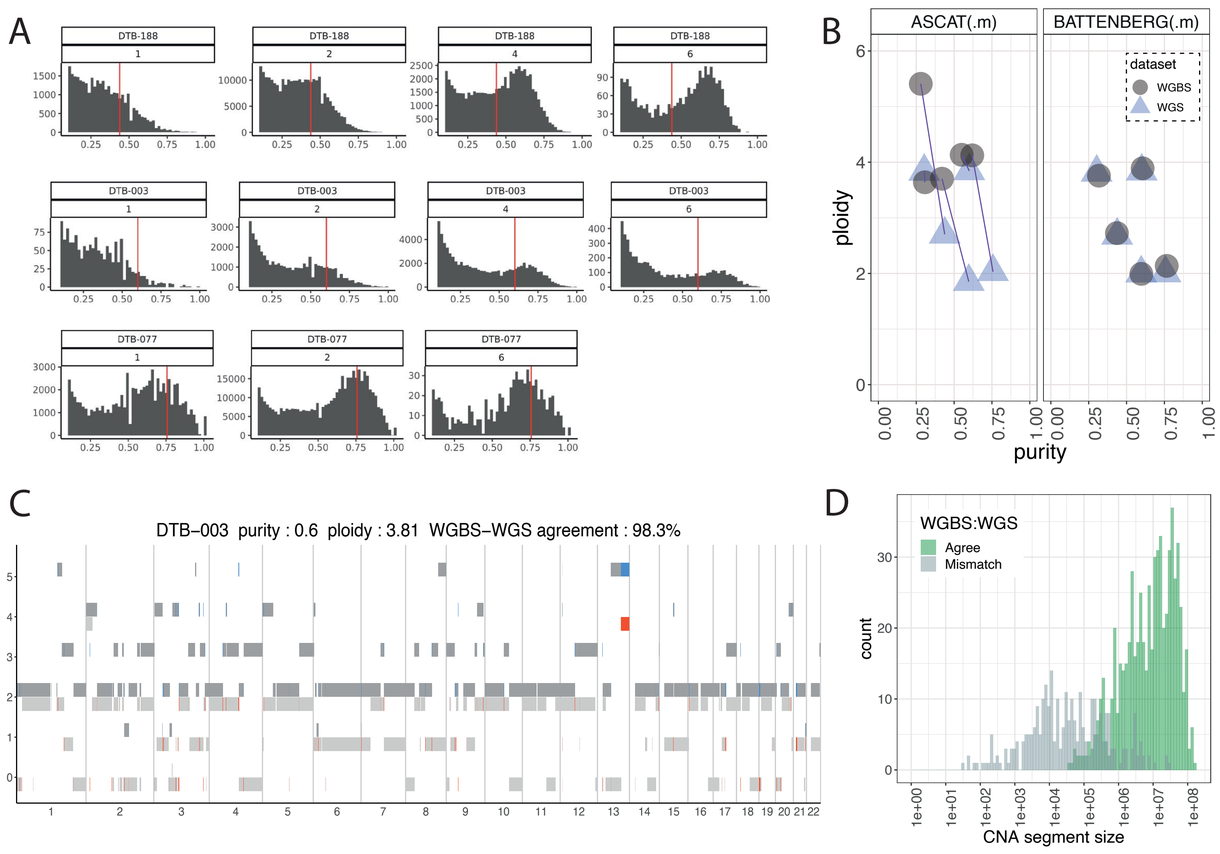

Figure 2. A) Bulk tumor methylation at CpGs unmethylated in matched normal tissue. Red line: purity estimates from WGS. B) Purity and ploidy estimates from applying CAMDAC to WGBS data alongside estimates from WGS. C) Allele-specific copy number concordance for sample DTB-003 at CNA. Dark grey: Matching major allele, Grey: matching minor allele, Blue: Major allele mismatch, Red: Minor allele mismatch. D) Allele-specific copy number abberation (CNA) concordance at overlapping CNA segments between all tumor samples.

Pure tumor methylomes cluster by tumor subtype

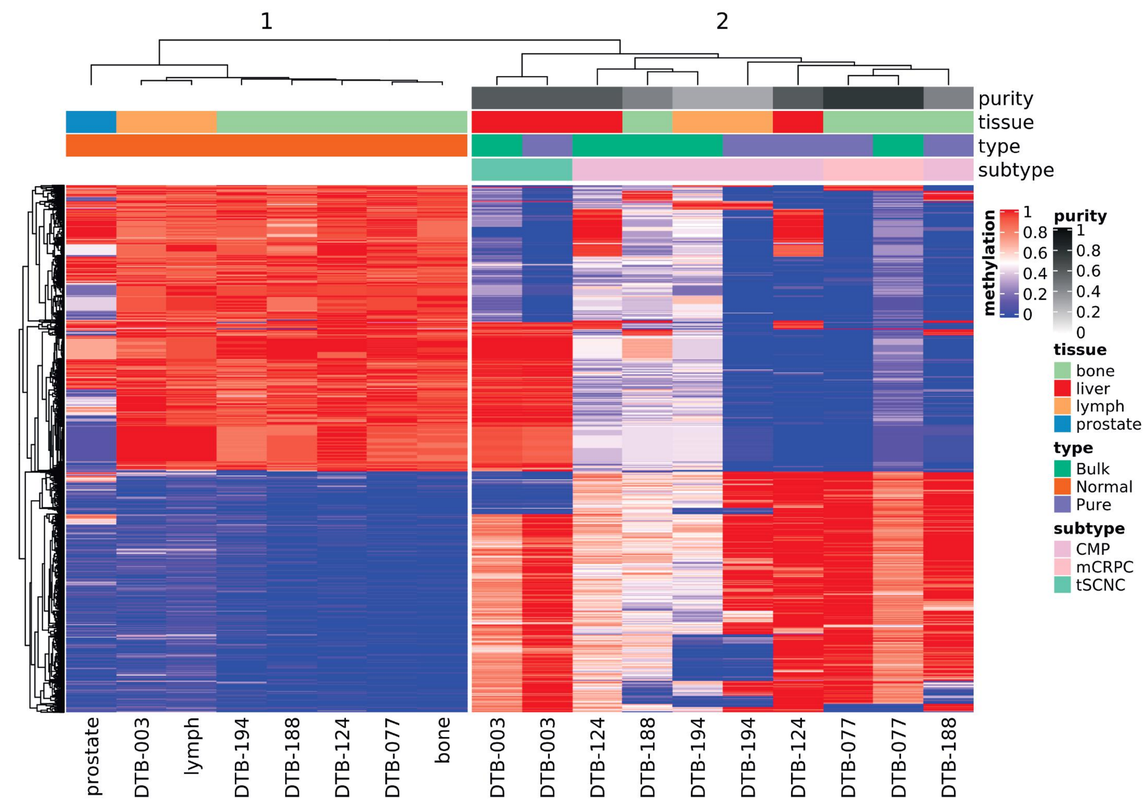

Figure 4. A) Hierarchical clustering of DNA methylation profiles from bulk tumor, normal, purified tumor and normal panel samples. Normal panels group with patient-matched normals (cluster 1). Bulk and purified tumor samples group together (cluster 2) with bulk tumors displaying more intermediate methylation, indiciating greater cellular admixture. Once deconvolved, pure tumors cluster separately from bulk tumors for the same sample, except in the case of DTB-077 due to high purity, and DTB-003 driven by its identity as a neuroendocrine tumour.

CAMDAC deconvolution reproduces SNV-phased methylation

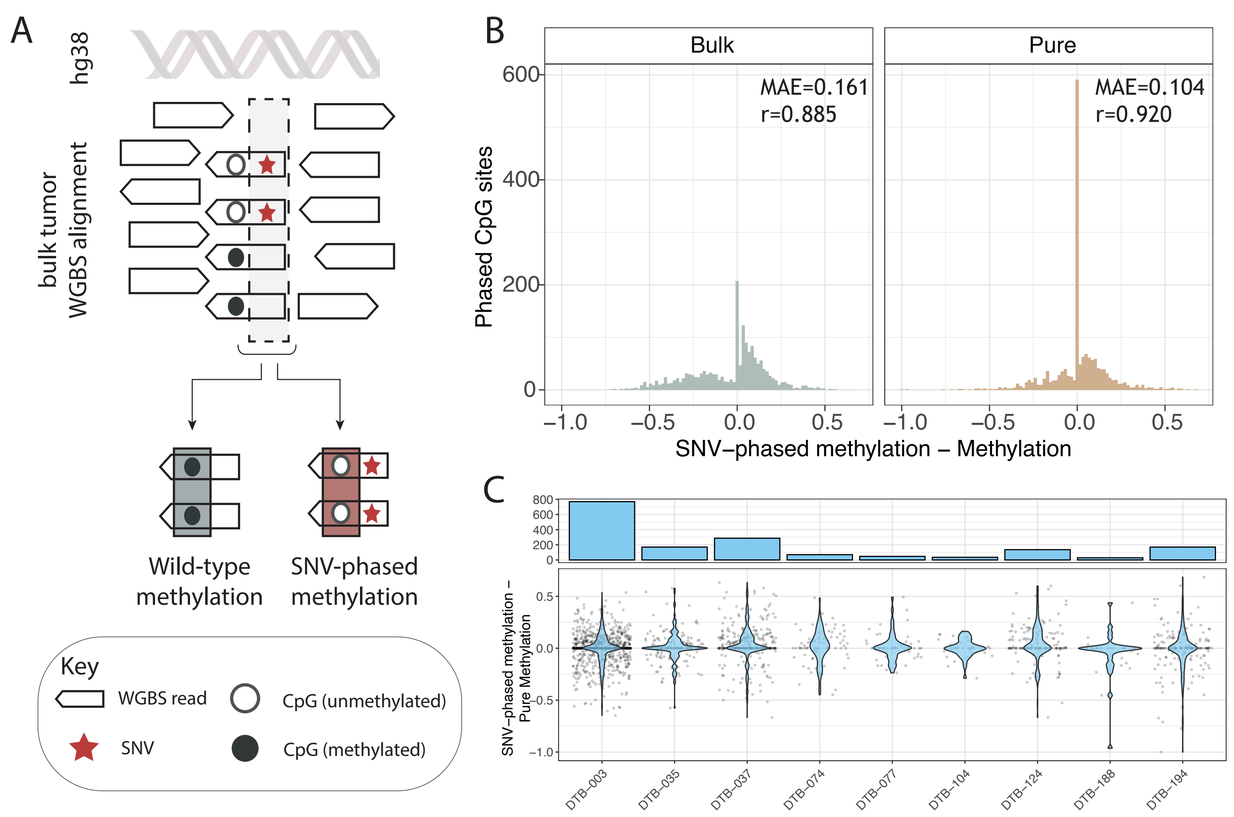

Figure 3. A) A schematic of the procedure for CAMDAC validation using methylation on SNV-phased reads. B) Bulk and pure methylation at sites with SNV-phasing. MAE: median aboslute error C) Per-sample distribution of CAMDAC-purified methylation distance from SNV-phased methylation at phased CpGs. Barplots (top) display the number of SNV-phased CpGs evaluated for each sample.

Normal methylation panels are similar to patient-matched tissue-matched samples

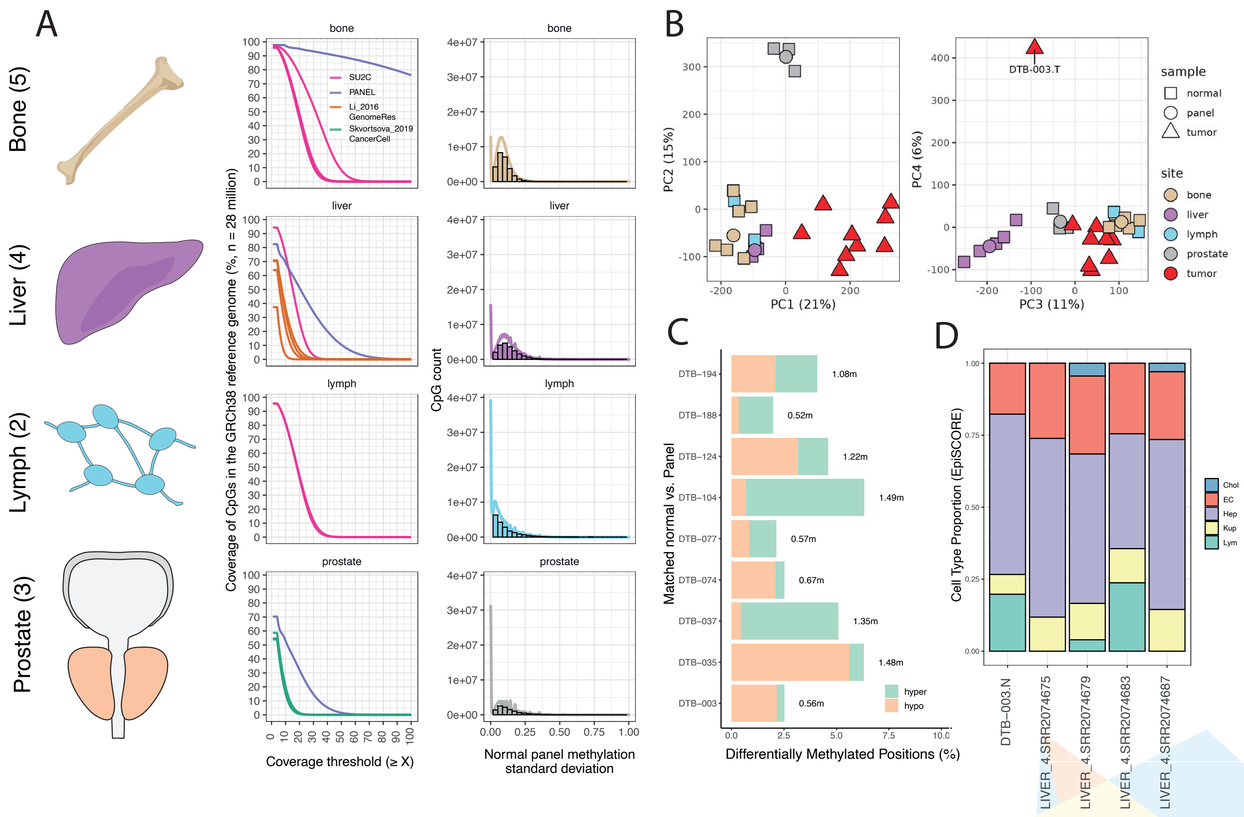

Figure 5. A) CpG coverage and standard deviation for each normal tissue methylation panel. B) Principal components analysis reveals sample methylation variance is explained by tumor-normal differences (PC1), prostate samples (PC2), liver samples (PC3) and neuroendocrine tumors (PC4). C) Differential methylation between patient matched normals and a tissue-matched validation panel. D) Cell type proportions inferred across normal liver samples using EpiSCORE.

Conclusions

CAMDAC estimates pure tumor methylation from bulk WGBS data, as demonstrated with the mCRPC dataset. SNV-phased reads demonstrate that CAMDAC-deconvolved methylation rates are superior to bulk methylation. Normal tissue methylation panels may be suitable for deconvolution, dependent on inter-sample cell type composition. Future work will profile the interplay between CNAs and epigenetic disregulation. CAMDAC is available at https://github.com/VanLoo-lab/CAMDAC/

References
1. Zhao, S.G., et al., 2020. The DNA methylation landscape of advanced prostate cancer. *Nature genetics*, 52(8), pp.778-789
2. Larose Cadieux, E., et al., 2022. Copy number-aware deconvolution of tumor-normal DNA methylation profiles. *bioRxiv*.